Physical Adversarial Background Patch Against Aerial Object Detection Based on Pareto Efficiency

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***Abstract*— For adversarial attacks on aerial image object detection, some physical background attack methods have been proposed and demonstrated excellent performance. However, most of the methods restrict the protected object to a fixed area, and changes in the position and size of the object can affect the effectiveness of the attack. In addition, the size of the background is only related to the size of the object, without considering any reduction in the background area. To alleviate these issues, this article proposes a novel adversarial background patch generation strategy, which generates an adversarial background patch where aircrafts parked at any position within it remain undetected. Besides, we also aim to make the size of the background as small as possible to improve the concealment of the attack and reduce the economic cost. Specifically, a series of adversarial images is generated by placing the aircraft at random angles and positions on the adversarial background patch. Then, a novel background patch optimization strategy is proposed, which enhances the occlusion robustness of the adversarial background patch by weighting the adversarial loss of each image in the batch based on adversarial difficulty. In addition, this article proposes a novel area loss for achieving optimal area size, which is used to reduce the cost of producing the patch. Finally, a multiobjective optimization method based on Pareto efficiency is introduced for balancing two conflicting losses, the adversarial loss and the area loss. Experimental results show that the adversarial background patches generated by this method have excellent attack effects in both digital and physical attacks, exhibiting strong occlusion robustness. In addition to that, the adversarial background patches generated by this method achieve effective attacks in a smaller area and reduce the physical implementation cost.**

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I. Introduction

With the rapid development of deep neural networks (DNNs), DNN-based object detectors are widely used in real-world applications such as intelligent surveillance [1], autonomous driving [2], and others [3], [4]. However, recent studies have indicated that DNNs are shown to be highly vulnerable to adversarial examples, i.e., images added with intentionally designed yet imperceptible perturbations that are usually unnoticed by the human visual system but can fool DNNs. On the one hand, it raises concerns about the security of DNN models when deployed in real-world scenarios. On the other hand, the adversarial attack methods provide more insights for improving the adversarial robustness of DNNs.

Aerial images usually contain a large amount of geographic information, which is crucial for applications such as environmental monitoring [5], urban planning [6], disaster assessment [7], and others [8], [9], [10]. Object detectors based on DNNs, with the capability to handle both classification and localization simultaneously, hold a pivotal role in the understanding and analysis of aerial images. In general, object detectors of aerial images can be divided into two-stage detectors, single-stage detectors, and others [11]. For the two-stage detectors, Cheng et al. [12] suggested to learn a rotation-invariant CNN model for multiclass geographic object detection. By incorporating multiangle anchors into the current region proposal network (RPN) based on the Faster R-CNN pipeline, a rotation-insensitive RPN [13] was developed to effectively address the issue of geographical object rotation changes. In order to achieve real-time object detection, a few studies exploited one-stage detectors on remote sensing images, such as YOLO series and others [11]. Liang et al. [14] propose an enhanced feature fusion and scaling-based single-shot detector method for small object detection in UAV images, featuring improved feature representation and spatial context analysis, which achieves superior accuracy compared to state-of-the-art methods while maintaining comparable detection speeds. Furthermore, transformer-based detectors have garnered significant interest due to their superior capability for high-precision detection [15]. An aspect ratio-sensitive detection transformer was proposed in [16] for oriented object detection in aerial images, which includes an

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aspect ratio-aware circle smooth label strategy and a rotated deformable attention module to eliminate the misalignment between region features and sampling points.

Unfortunately, there still exist vulnerabilities in the object detection of aerial images. Common adversarial attack methods add perturbations based on model gradient optimization to the image to obtain adversarial samples [17], [18]. Tang et al. [19] proposed a natural weather-style adversarial attacks method to produce more natural and stealthy adversarial examples. Besides, Sun et al. [20] proposed a novel attack method called task-specific importance aware attack for targeting object detection in optical remote sensing images, which employs soft-weighted ensemble optimization to enhance attack efficiency and significantly improves the success rate in targeted attack tasks. These attack methods perform well in the digital world, but are challenging to implement in the physical world due to the difficulty of perturbations being captured by optical cameras. In order to make adversarial examples useful in real-world situations, some recent studies [21], [22] have proposed the adversarial patch to induce object detectors to misclassify targets or make protected objects invisible by changing the texture [23] or shape [24] of patches. In order to solve the problem of generating universal adversarial perturbations for multiscale objects, Zhang et al. [25] add a scale factor on the basis of adversarial attacks to generate universal adversarial patches that adapt to multiscale objects and ensure the effectiveness of adversarial patches. Moreover, to improve the concealment of the patch, style transfer was introduced to generate adversarial patches with natural textures [26], which is difficult for humans to identify.

Most of the existing physical patch attack methods [24], [25], [26], [27] use patches to be placed directly above the object, which may cause serious occlusion to the object, and each object needs a patch with a suitable scale, making the patch highly non-reusable. In order to solve the above problems, a context background attack method was proposed in [28], which extracts the saliency of the object of interest as the mask of the adversarial patch, and optimizes the area outside the mask area to closely cover the key context background area for detection. Wang et al. [29] proposed an unocclusion training strategy, which places the adversarial background under the object while optimizing the pixel values of the background outside the target area. The adversarial background can conceal the chosen objects and lead to the adversarial pattern in the background being incorrectly classified as belonging to a particular category. Zhang et al. [30] proposed an adversarial textures generation method, which utilized the translational invariance of fully convolutional networks to decouple adversarial textures from shapes, allowing adversarial textures to be arbitrarily expanded during use. However, they inevitably exhibit certain limitations. On the one hand, the position of the protected object is limited to a certain area. Tiny changes in the angle and position of the protected object may make the designed physical adversarial background patch ineffective. On the other hand, in order to achieve the superior stealthiness and low economic cost, the size of the adversarial background patch is expected to be as small as possible when it is deployed in the physical world.

In order to alleviate the above issues, a novel physical adversarial background patch generation method is proposed to fool object detectors, which places the object on the generated background patch at any position and angle. First, a series of adversarial images is obtained by placing the random angle and position of the protected object on the adversarial background patch. Then, a batch optimization strategy is designed to achieve an effective attack on all adversarial images. The proposed strategy balances the aggressiveness and robustness of the adversarial background patch by evaluating the attack difficulty of each adversarial image and calculating the corresponding weight. Second, the adversarial background patch is expected to be as small as possible when it is deployed in the physical world. To this end, a novel area loss is constructed to balance the attack aggressiveness and physical implementation cost. Finally, a multiobjective optimization method based on Pareto efficiency is introduced to obtain Pareto efficient solutions. Compared with other multiobjective optimization methods, the Pareto-efficient-based multiobjective optimization method can adaptively adjust the loss weights to avoid sacrificing one objective while optimizing another, and the degradation of optimization due to manually assigning weights. For a comprehensive evaluation, both black-box and white-box attacks are tested in both the digital and physical domains using one-stage detectors, two-stage detectors, and transformer-based detectors. The results demonstrate the effectiveness and robustness of the generated adversarial background patch for both digital and physical attacks. In summary, our contributions are as follows.

1) To enhance the occlusion robustness of the adversarial patches, this article proposes a physical adversarial background patch, and the object can be positioned at any angle above the patch. An occlusion robustness enhancement method based on batch optimization is proposed to generate a series of adversarial samples, which are associated with dynamic weights measured by difficulty evaluation.
2) To achieve the superior stealthiness and low economic cost, a multiobjective adversarial attack model is established with an adversarial loss and an area loss to balance the attack aggressiveness and the perturbation area. In addition, a multiobjective optimization method based on Pareto efficiency is used to balance these two conflicting objectives.

The rest of this article is organized as follows. Section II reviews the related work of adversarial attacks on object detection. In Section III, the details of the adversarial background patch generation method are presented. In Section IV, the effectiveness of the adversarial background patch is demonstrated through a series of experiments. Section V summarizes the proposed method and proposes potential future works.

II. Related Work and Motivation

In this section, the related works of adversarial attacks in the field of object detection are briefly introduced. Next, recent research on adversarial attacks in aerial image object detection is reviewed.

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A. Adversarial Attacks

In general, adversarial attacks can be divided into digital attacks and physical attacks according to the domain of the attack. A digital attack generates adversarial samples by directly tampering with the pixel values of clean samples. Typical methods of digital attack include FGSM [31], BIM [32], PGD [33], C&W algorithm [34], and deepfool [35]. Different from object vanishing adversarial attack [36], [37], [38], Li et al. [39] proposed ShiftAttack, which realizes feature exchange across different regions by applying imperceptible perturbations to images, thereby moving the predicted bounding boxes of any target in arbitrary directions and distances. The methods mentioned above are all white-box attacks, that is, the structure and parameter information of the victimized model can be obtained. However, the perturbation generated by these methods is ineffective in the physical world, for the reason that small noises are difficult to capture by physical optical sensors. As a result, physical attack techniques are beginning to take notice. Chen et al. [40] propose ShapeShifter and demonstrate that the EOT technique, which was first put forward to strengthen the resistance to adversarial perturbations in image classification, can be successfully extended to the object detection domain. To apply strong physical attacks against DNN-based person re-identification systems, Wang et al. [41] proposed advPattern to create adversarial patches on clothing that can prevent persons from being detected. In addition to the visible light field, by modeling the transition from fabric to clothes in the digital domain and creating an adversarial "QR code" pattern, Zhu et al. [42] proposed the infrared adversarial clothing. Unlike adversarial attack methods that invalidate the victim model, Sun et al. [43] proposed an improved method for black-box model stealing, which uses a joint data optimization scheme and self-conditioned data synthesis framework to enhance the training efficiency and effectiveness of proxy models.

B. Physical Adversarial Attacks on Aerial Image Object Detection

Currently, due to the wide application of aerial image object detectors in various scenes [44], the adversarial attack methods against them have attracted attention. In terms of physical implementation, Du et al. [45] designed a method to significantly reduce the effectiveness of aerial object detectors by optimizing, manufacturing, and installing adversarial patches on or near the target objects. Additionally, Lian et al. [27] proposed a framework called adaptive patch-based physical attack, aiming to generate adaptive adversarial patches. By designing new loss functions to consider more available information for optimizing adversarial patches, the attack effectiveness was significantly improved. Furthermore, a contextual background attack [28] was proposed, which introduced a novel framework for contextual background attacks, targeting aerial detectors in the physical world. This framework uses target objects of interest (such as aircrafts) in aerial images to mask adversarial patches. Then, by optimizing pixels outside the masking area, the generated adversarial patches tightly cover key contextual background areas used for detection, thereby imparting more robust and transferable attack capabilities to the adversarial patches. Wang et al. [29] proposed an unoccluded training strategy, which makes use of an image's target mask to strategically place adaptive adversarial background beneath the targets while updating and optimizing the background's pixel values outside of the target region. In addition, a dual deceptive loss function was introduced to achieve both targeted and untargeted attacks.

III. Methodology

In this section, the definition of the adversarial background patch is shown first. Then, the overall framework of designing a physical adversarial background patch is proposed to attack aerial image object detectors. Finally, the loss design is introduced, and a multiobjective optimization method based on Pareto efficiency is used to balance losses of conflict.

A. Problem Definition

Physical adversarial background patch aims to make the protected object, which is placed above the patch at an arbitrary angle and position, undetected by the object detectors. Specifically, let us consider a benign aerial image $\boldsymbol{x}$ and a protected target $\boldsymbol{T}$. Take the aircraft as an example, the aircraft is randomly scaled and rotated randomly on the adversarial background patch $\boldsymbol{P}$ to obtain the adversarial image $\boldsymbol{P}^*$. Finally, the adversarial example $\boldsymbol{x}^*$ with the adversarial background patch $\boldsymbol{P}$ can be formulated as

$$\boldsymbol{x}^* = \boldsymbol{x} \odot (1 - \boldsymbol{M}_{\boldsymbol{P}^*}) + \boldsymbol{P}^* \odot \boldsymbol{M}_{\boldsymbol{P}^*} \tag{1}$$

where $\odot$ represents Hadamard product, $\boldsymbol{M}_{\boldsymbol{P}^*}$ represents the mask of adversarial image $\boldsymbol{P}^*$, and the pixel values of the foreground are 1 and the rest are 0. The proposed adversarial background patch has the following two characteristics.

1) Improve Occlusion Robustness of the Patch: Due to the occlusion between the object and the patch, small adjustments to the object's location and angle could make the physical adversarial background patch useless. In this article, a batch optimization strategy is proposed to improve occlusion robustness of the adversarial background patches by evaluating the attack difficulty of each adversarial image and calculating the corresponding weight.

2) Improve Concealment and Reduce Economic Costs: When the adversarial background patch is implemented in the real world, its size is anticipated to be as tiny as feasible in order to improve stealthiness and reduce economic cost. Area loss is introduced to balance the attack aggressiveness, the concealment of the patch, and the physical implementation cost. However, attack effectiveness and patch area are two conflicting objectives, which are difficult to optimize simultaneously. In order to alleviate this issue, a gradient descent strategy based on Pareto efficiency is used, which can find the Pareto optimal solution of multiple conflicting objectives.

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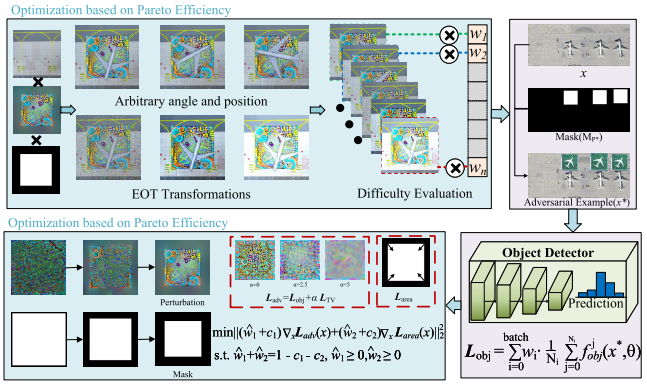


Fig. 1. Framework of the adversarial background patch generation.

B. Physical Adversarial Background Patch Generation

In order to improve the attack and robustness of patches in the physical world, an adversarial background patch generation strategy is proposed, which can be categorized into two main parts focusing on data preparation and optimization. In the first part, the adversarial samples for training are generated by different occlusion and physical data augmentation methods. In the second part, a batch optimization strategy is proposed to implement a universal attack on all adversarial samples. The framework for generating specific adversarial background patches is shown in Fig. 1.

1) Adversarial Example Generation: For generating the adversarial background patch, an aerial image without any object is given as the initial background image $\boldsymbol{p_0}$, and then the angle of the protected object (aircraft in this article) $\boldsymbol{T}$ is randomly sampled from 0° to 360°, and placed in the background image with a random size and position to obtain adversarial image $\boldsymbol{p^*}$.

After that, the adversarial image is scaled to the approximate size of the aircraft in the aerial image and placed next to it, so that the aircraft inside and outside the adversarial background patch can be detected, which helps to improve the attack effect of the adversarial background patch. Specifically, for the label information (such as YOLO labels) containing aircrafts in the clean image, including the coordinates of the top-left and bottom-right corners of the target (x_1, y_1, x_2, y_2) and the class, we calculate the position l_{P^*} and size (W, H) of the adversarial image using the following formula:

$$\boldsymbol{l_{P^*}} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} - \frac{y_2 - y_1}{r_d}\right) \tag{2}$$

$$W = H = \sqrt[2]{r_s \cdot (x_2 - x_1) \cdot (y_2 - y_1)} \tag{3}$$

where r_d and r_s represent the distance coefficient and size coefficient for adjusting the position and size of the embedded adversarial image, respectively.

EOT is a broad framework for enhancing the robustness of adversarial patching by considering a given transformation distribution during the optimization process, which can be defined as

$$\widetilde{\boldsymbol{p^*}} = \mathbb{E}_{d\sim D}\left(f\left(\boldsymbol{p^*} + d\right)\right) \tag{4}$$

where $\mathbb{E}_{d\sim D}$ represents the EOT transformation, and d is a transformation perturbation selected from the perturbation distribution $D = \{d_1, d_2, d_3, \ldots, d_n\}$, including illumination, contrast, and noise.

Based on the obtained $\boldsymbol{l_{p^*}}$, w, and h, the mask of adversarial image $\boldsymbol{M_{P^*}}$ and scale-adapted adversarial image can be defined as follows:

$$\left[\boldsymbol{P^*}, \boldsymbol{M_{P^*}}\right] = \boldsymbol{PA}\left(\boldsymbol{x}, \widetilde{\boldsymbol{p^*}}, \boldsymbol{l_{p^*}}, W, H\right). \tag{5}$$

Finally, the adversarial sample is generated according to formulation (1).

2) Batch Optimization Strategy: Based on the above patch generation strategy, a series of adversarial samples containing patches occluded at different angles is obtained. Due to the complexity and diversity of these adversarial samples, the average weight is difficult to judge the effectiveness of each sample perturbation. Therefore, a weight dynamic adjustment strategy is designed to measure the attack difficulty of each adversarial example. Specifically, the corresponding weight is calculated according to the adversarial loss of each adversarial sample, and then the patch is optimized by using the normalized weight.

Due to the large number of adversarial samples, adversarial perturbations are difficult to validate for all adversarial samples, i.e., the loss increases greatly after adding adversarial perturbation to some adversarial samples, while the loss remains essentially unchanged after adding adversarial perturbation to other samples, and thus difficulty evaluation weights of the perturbations w^i_{dif} are defined as

$$\boldsymbol{w}^{\boldsymbol{i}}_{\text{dif}} = \frac{E_i(r)}{E_1(r)} = \frac{\frac{1}{n_i}\sum_{j=1}^{n_i} f^j_{\text{obj}}(x_i, \theta)}{\frac{1}{n_1}\sum_{j=1}^{n_1} f^j_{\text{obj}}(x_1, \theta)} \quad i \in (1, 2, \ldots, N_{\text{batch}}) \tag{6}$$

where r represents the result of object detector recognition with model parameter θ, which generally includes the bounding boxes f_{pos}, the object probability f_{obj}, and the class score f_{cls}. $E(r)$ denotes the extraction of objective probability. n_i represents the number of objects identified in the image. N_{batch} represents the number of adversarial images in the batch.

Attack difficulty quantifies the ratio of the adversarial loss between the ith image and the first image in the batch. For example, when this ratio is relatively large, it indicates that the ith image suffers greater loss, making it more difficult for adversarial interference to attack, thus requiring an increase in the weight of this image. The weight for each image is calculated as follows:

$$\boldsymbol{w}^{\boldsymbol{i}}_{\text{dif}} = \frac{\exp\left(\boldsymbol{w}_{\text{dif}}{}^{i}\right)}{\sum_{j=1}^{N_{\text{batch}}} \exp\left(\boldsymbol{w}^{\boldsymbol{j}}_{\text{dif}}\right)} \tag{7}$$

thus, we obtain adaptive weight values for each image. Finally, object detection loss is represented as

$$\boldsymbol{L}_{\text{obj}} = \sum_{i=1}^{N_{\text{batch}}} \boldsymbol{w}^{\boldsymbol{i}}_{\text{dif}} \cdot \frac{1}{n_i} \sum_{j=1}^{n_i} f^j_{\text{obj}}\left(x^*_i, \theta\right). \tag{8}$$

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Algorithm 1 Adversarial Background Generation Algorithm

Input: Original background p_0, aircraft image T, victim detector F, learning rate δ, loss function L, the number of iterations epoch N_{epoch}, the number of batch N_{batch}, hyperparameter α
Output: The adversarial background patch p
Procedure:
Initialize the victim detector F
for $i = 1$ to N_{epoch} **do**
  for $j = 1$ to N_{batch} **do**
    Target T random rotation scaling is placed on the background to obtain p^*;
    $\widetilde{\boldsymbol{p^*}} = \mathbb{E}_{d\sim D}(f(\boldsymbol{p^*} + t))$;
    $[P^*, M_{P^*}] = PA(x, \widetilde{p^*}, l_{p^*}, W, H)$;
    $\boldsymbol{x^*} = \boldsymbol{x} \odot (1 - \boldsymbol{M_{P^*}}) + \boldsymbol{P^*} \odot \boldsymbol{M_{P^*}}$;
    $r = F(x^*)$;
    $L^j_{obj} = E(r)$;
  end for
  Calculate difficulty weight w^j_{dif};
  $L_{\text{obj}} = \sum_{j=1}^{N_{\text{batch}}} w^j_{\text{dif}} * L^j_{\text{obj}}$;
  $L_{\text{adv}} = L_{\text{obj}} + \alpha L_{\text{TV}}$;
  Calculate the loss weight w_1, w_2, find the Pareto solution;
  $L_{\text{total}} = w_\Delta L_{\text{adv}} + w_\Theta L_{\text{area}}$;
  $p_i \leftarrow p_i - \delta \nabla_p L_{\text{total}}$;
end for
return p;

C. Multiobjective Optimization Based on Pareto Efficiency

In this article, we aim to reduce the economic costs of physical adversarial background patches to make the background perturbation area as small as possible. Therefore, the considered loss function comprises the adversarial loss and the area loss. Specifically, the adversarial loss consists of object detection loss and the total variation smoothness loss. Finally, these losses are minimized by a multiobjective optimization method based on Pareto efficiency.

1) Multiobjective Model Based on the Adversarial and Area Losses: To enhance the adversarial effect of the perturbation toward the aircraft class, we introduce an object detection loss with batch optimization. This loss is based on the confidence of all aircraft targets in the image, as shown in (8).

1) *Adversarial loss:* In physical-world attacks, cameras often struggle to capture details of adversarial perturbation, which may result in degraded or even ineffective performance of adversarial perturbation. To address this issue, we introduce TV smoothness loss, which can be defined as

$$\boldsymbol{L}_{\text{TV}}(p) = \sum_{i,j}\left[\left(p_{i,j} - p_{i+1,j}\right)^2 + \left(p_{i,j} - p_{i,j+1}\right)^2\right] \tag{9}$$

where $P_{i,j}$ represents the pixel at the ith row and jth column of P. The TV smoothness loss can make the generated adversarial perturbation smooth at the pixel level, reduce noise and discontinuity, and improve its effectiveness in the physical world. In summary, the adversarial loss can be defined as the following formula:

$$\boldsymbol{L}_{\text{adv}} = \boldsymbol{L}_{\text{obj}} + \alpha \boldsymbol{L}_{\text{TV}} \tag{10}$$

where α is used to control the smoothness level.

2) *Area loss:* For physical attacks, concealment and economic costs are critical but easily overlooked. Hence, the area loss is adopted to balance aggressiveness and the area of the perturbation. Specifically, the size of the adversarial background patch $\boldsymbol{P}$ is defined as $\boldsymbol{A_P}$, the size of the initial background is $\boldsymbol{A_{P_0}}$, and the area loss $\boldsymbol{L}_{\text{area}}$ can be expressed by the following formula:

$$\boldsymbol{L}_{\text{area}} = \frac{\exp\left(\frac{\boldsymbol{A_P}}{\boldsymbol{A_{P_0}}}\right) - \exp\left(-\frac{\boldsymbol{A_P}}{\boldsymbol{A_{P_0}}}\right)}{\exp\left(\frac{\boldsymbol{A_P}}{\boldsymbol{A_{P_0}}}\right) + \exp\left(-\frac{\boldsymbol{A_P}}{\boldsymbol{A_{P_0}}}\right)}. \tag{11}$$

The hyperbolic tangent function is used to construct the loss, and the ratio of the two areas is transformed into a nonlinear function, which makes it change more smoothly and not too extreme in the optimization process.

2) Optimization Based on Pareto Efficiency: Considering the importance of attack effectiveness, in the training strategy, the area loss is adopted when all adversarial images in the batch are successfully attacked. In summary, the overall loss can be described by the following formula:

$$\min\{\boldsymbol{L}_{\text{adv}}, \boldsymbol{L}_{\text{area}}\}. \tag{12}$$

During the iterative process of generating adversarial perturbation, the effectiveness of the attack is in conflict with the size of the perturbation area, which may lead to a decrease in adversarial loss $\boldsymbol{L}_{\text{adv}}$ and an increase in area loss $\boldsymbol{L}_{\text{area}}$. To alleviate this issue, the concept of Pareto efficiency in multiobjective optimization is introduced to ensure that minimizing one objective does not harm any other objectives. Given a series of objective functions $\boldsymbol{L}_1, \boldsymbol{L}_2, \ldots, \boldsymbol{L}_n$, for two solutions $s_i = (\boldsymbol{L}^i_1, \boldsymbol{L}^i_2, \ldots, \boldsymbol{L}^i_n)$ and $s_j = (\boldsymbol{L}^j_1, \boldsymbol{L}^j_2, \ldots, \boldsymbol{L}^j_n)$, if every objective function in s_i is smaller than that in s_j, i.e., $\boldsymbol{L}^i_t < \boldsymbol{L}^j_t, \forall t \in n$, then s_i dominates s_j. When there is no other solution that dominates the solution $s_i = (\boldsymbol{L}^i_1, \boldsymbol{L}^i_2, \ldots, \boldsymbol{L}^i_n)$, it is called Pareto efficiency. Pareto-efficient solutions are not unique, and the set of all such solutions is called the Pareto front. Based on the above definitions, we can ensure that every loss function decreases by only seeking Pareto efficiency in each iteration.

For the aforementioned adversarial loss $\boldsymbol{L}_{\text{adv}}$ and area loss $\boldsymbol{L}_{\text{area}}$, these two losses are combined into one loss using scalarization techniques, i.e., $\boldsymbol{L}_{\text{total}} = w_1 \boldsymbol{L}_{\text{adv}} + w_2 \boldsymbol{L}_{\text{area}}$. To obtain Pareto-efficient solutions, we need to minimize the total loss function. Considering the Karush–Kuhn–Tucker (KKT) [46] conditions $w_1 + w_2 = 1, \exists w_i > c_i, i \in \{1, 2\}$, and $w_1 \nabla_x \boldsymbol{L}_{\text{adv}}(x) + w_2 \nabla_x \boldsymbol{L}_{\text{area}}(x) = 0$, the following optimization problem can be derived:

$$\begin{aligned} \min \quad & \|w_1 \nabla_x \boldsymbol{L}_{\text{adv}}(x) + w_2 \nabla_x \boldsymbol{L}_{\text{area}}(x)\|_2^2 \\ \text{s.t.} \quad & w_1 + w_2 = 1, w_1 \geq c_1, w_2 \geq c_2 \end{aligned} \tag{13}$$

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where c_1 and c_2 are constants between 0 and 1, and $c_1 + c_2 \leq 1$. The constraints c_1 and c_2 added to the objectives are predefined boundary constraints. It has been proven [47] that the solutions obtained from the above optimization problem either satisfy the KKT conditions or result in decreasing all the loss functions along the gradient direction. If a solution satisfies the KKT conditions, then it is Pareto-optimal.

Define $\hat{w}_1 = w_1 - c_1$ and $\hat{w}_2 = w_2 - c_2$, the above optimization problem can be reformulated as a quadratic programming problem

$$\begin{aligned} \min \ & \|(\hat{w}_1 + c_1)\nabla_x \boldsymbol{L}_{\text{adv}}(x) + (\hat{w}_2 + c_2)\nabla_x \boldsymbol{L}_{\text{area}}(x)\|_2^2 \\ \text{s.t.} \ & \hat{w}_1 + \hat{w}_2 = 1 - c_1 - c_2, \hat{w}_1 \geq 0, \hat{w}_2 \geq 0. \end{aligned} \tag{14}$$

To solve the above quadratic programming problem, we consider relaxing the problem by considering only equality constraints as follows:

$$\begin{aligned} \min \ & \|(\hat{w}_1 + c_1)\nabla_x \boldsymbol{L}_{\text{adv}}(x) + (\hat{w}_2 + c_2)\nabla_x \boldsymbol{L}_{\text{area}}(x)\|_2^2 \\ \text{s.t.} \ & \hat{w}_1 + \hat{w}_2 = 1 - c_1 - c_2. \end{aligned} \tag{15}$$

According to the paper [48], the solution to the above equality-constrained problem is given by $\begin{bmatrix} \widetilde{w}_1 \\ \widetilde{w}_2 \end{bmatrix} = ((M^T M)^{-1} M\widetilde{z})$, where M is the matrix $\begin{bmatrix} GG^{\mathrm{T}} & e \\ e^{\mathrm{T}} & 0 \end{bmatrix}$, G is the stacking matrix of $\boldsymbol{L}_{\text{adv}}$ and $\boldsymbol{L}_{\text{area}}$, e is the vector whose elements are all 1, c is the concatenated vector of c_1 and c_2, and $\widetilde{z}$ is the concatenated vector of $-GG^T c$ and $1 - c_1 - c_2$.

Since the non-negativity constraints are ignored, the obtained solution may be invalid. To ensure the validity of the solution, the following projection step is performed:

$$\begin{aligned} \min \ & \left[(\widetilde{w}_1 - \hat{w}_1^*)^2 + (\widetilde{w}_2 - \hat{w}_2^*)^2\right] \\ \text{s.t.} \ & \widetilde{w}_1 + \widetilde{w}_2 = 1, \widetilde{w}_1 \geq 0, \widetilde{w}_2 \geq 0. \end{aligned} \tag{16}$$

The problem is a non-negative least-squares problem, which can be easily solved using the active set method [49].

D. Analysis on Physical Adversarial Background Patch

In the design of adversarial background patches, it is of great significance to optimize the confrontation effect and area at the same time. First of all, by minimizing the interference area, the influence of the background on the human visual system can be reduced, making the background more subtle and less detectable. Moreover, the smaller perturbation area means that it can be applied in a more diverse environment without significantly changing the original background, so it has better adaptability and flexibility. Last but not least, reducing the material or display area required to counter the background can reduce the economic cost.

In the implementation of adversarial attacks in the physical world, the adversarial background patches we generate also have advantages in cost-effectiveness and flexibility. On the one hand, the area constraint reduces the economic cost of the patch in actual production, and the robustness of the adversarial background patch against different objects makes it have good reusability. On the other hand, the improvement of the occlusion robustness of the adversarial background patch makes it simple to the implement. One background can attack objects of different scales, and the object does not need to be deliberately parked in a certain position or angle, which greatly improves the flexibility.

IV. Experimental Study

In this part, a comprehensive experimental study is carried out to prove the effectiveness of the proposed adversarial background patch in the digital domain and the physical world. Then, the ablation experiments of TV smooth loss and batch optimization were carried out to verify its effectiveness.

A. Experimental Settings

1) Datasets: Two commonly used large-scale remote sensing datasets are used for training object detectors and adversarial background patch generation, i.e., DOTA [55] and RSOD [56], [57]. The DOTA dataset contains 2806 samples with image sizes ranging from 800 × 800 to 4000 × 4000. It contains 15 categories, including aircraft, ships, cars, and other commonly used man-made objects. The RSOD dataset contains four categories of aircraft, fuel tanks, playgrounds, and overpasses, which contain about 5000 aircrafts.

2) Victim Model: Ten object detectors are selected as the victim models for our study, encompassing a wide range of architectures: from the two-stage detector Faster R-CNN [50], to the one-stage detectors SSD and multiple iterations of the YOLO family (including YOLOv3 [52], YOLOv5 [53] in its various sizes such as n, s, m, l, x, and YOLOv8n [54]), and finally, the transformer-based detector DETR [11]. Specifically, both Faster R-CNN and DETR use resnet-50 as a backbone network. The detailed training results of the victim model are shown in Table I.

3) Compared Methods: Two adversarial patch attack methods and a background attack method are selected for comparison, including adversarial patches proposed by Thys et al. [58], APPA proposed by Lian et al. [27], and CBA [28] attack method.

4) Implementation Details: During the training process of adversarial background patch, α is adjusted to 2.5 to balance the objectiveness loss and the overall variation. The iterative optimization process, the threshold of the intersection-union ratio (IOU), and the objective confidence of the object detector were set to 0.45 and 0.4. The optimizer uses the Adam algorithm, and the initial learning rate lr is set to 0.3. All the codes are implemented in PyTorch on RTX4090 (24 GB) GPUs.

B. Evaluation Metrics

1) Attack Success Rate (ASR): In this article, the definition of ASR is divided into two aspects.

1) Class attack success, defined as the class failure of the object detector to predict the target. Specifically, targets recognized by the object detector as other classes as well as those with confidence less than 0.2 will be considered as attack success, which can be defined as $\phi_{\text{class}}(t) = (\hat{c}(t) \neq c(t) \vee \text{conf}(t) < 0.2)$.
2) Bounding box attack success, defined as the bounding box predicted by the object detector to the bounding

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TABLE I

METRIC (MAP@.5:.95) OF VARIOUS MODELS ACROSS DOTA AND RSOD ARE PRESENTED, WHERE YOLOV3 HAS BEEN IMPROVED UPON THE ORIGINAL STRUCTURE WITHIN [52], FASTER R-CNN USES RESNET50 AS BACKBONE

Model	DOTA	RSOD
Faster-RCNN [50]	.524	.490
SSD [51]	.531	.476
YOLOv3 [52]	.491	.432
YOLOv5s [53]	.480	.374
YOLOv5n [53]	.505	.432
YOLOv5m [53]	.544	.421
YOLOv5l [54]	.512	.453
YOLOv5x [54]	.550	.498
YOLOv8n [54]	.524	.509
DETR [11]	.542	.520

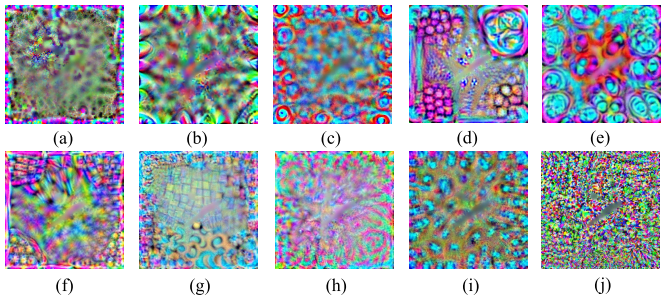


Fig. 2. Adversarial background patches generated against different object detectors. The patch used is based on YOLOv5n: (a) Faster R-CNN, (b) SSD, (c) YOLOv3n, (d) YOLOv5n, (e) YOLOv5s, (f) YOLOv5m, (g) YOLOv5l, (h) YOLOv5x, (i) YOLOv8n, and (j) DETR.

box in the original image with IoU less than 0.4, which can be defined as $\phi_{\text{bbox}}(t) = (\text{IoU}(\hat{B}(t), B(t)) < 0.4)$. The total ASR combines the above two aspects and is defined as the following formula:

$$\text{ASR} = \frac{\text{num}(\{t \in T \mid \phi_{\text{class}}(t) \vee \phi_{\text{bbox}}(t)\})}{\text{num}(T)} \tag{17}$$

where T is the set of all original targets in the datasets. $\hat{c}(t)$ is the predicted class label for target t. $c(t)$ is the ground truth class label. $\text{conf}(t)$ is the confidence score. $\hat{B}(t)$ is the predicted bounding box, and $B(t)$ is the ground-truth bounding box.

C. Digital Attack

For digital attacks, the attack dataset is created, which extracts the images containing the aircraft from the DOTA dataset and the RSOD dataset, and performs image segmentation on the aircraft to generate a mask file. Finally, according to the mask file and label data, the adversarial background patch is adjusted to the target size and placed under the target aircraft to generate an adversarial example.

The adversarial background patches generated for different object detectors are shown in Fig. 2. The ASR and transferability of the adversarial background patches are shown in Table II. Here are the key observations.

1) *Observation 1:* For different size models with the same structure, such as YOLOv5s, YOLOv5n, YOLOv5m, YOLOv5l, YOLOv5x, the generated adversarial background patches have similarities in terms of texture styles and color tones. Besides, the same series of models, such as YOLOv3 and YOLOv8, also have certain similarities. The two-stage detector Faster R-CNN has a significant difference in texture details compared to the single-stage detectors YOLO and SSD, and does not produce significant chunks of texture. It is worth noting that the adversarial background patches generated for DETR are closer to noisy images, making it difficult to optimize for obvious textures.
2) *Observation 2:* On the attack effect of adversarial background patches. In the same structure model (e.g., YOLO series), the adversarial background patches generated against YOLOv5s and YOLOv5m show, in addition to excellent attack effectiveness, transferability. In addition, too large or too small parameters of the model can affect the attack transferability; for example, the adversarial background patches generated for YOLOv5n, YOLOv3n, and YOLOv8n have poor transferability. Attacks against Faster R-CNN and DETR have lower success rates compared to YOLO and SSD. Meanwhile, the adversarial background patches generated against Faster R-CNN and DETR have poor transferability, which may be related to the model structure and principle.

D. Comparison With SoTA Methods

Note that in the comparative experiment of the digital domain, the specific implementation of the method is different. As shown in Fig. 3, adversarial patches proposed by Thys et al. [58] were placed on the target aircraft. In AP-PA [27], patches are placed outside the target. The digital experiment of CBA [28] places the generated adversarial patches next to the target. For BADEI [29], Adv Texture [30], and our method, the antagonistic background is placed below the target. The experimental results are shown in Table III. Here are the key observations.

1) *Observation 1:* Our method, which places the adversarial background patch under the target, demonstrates a significant advantage over other methods that place the patch on or outside the target (Thys, APPA, and CBA). For example, on YOLOv5s, our method achieves a 95.3% success rate in white-box attacks, while Thys, APPA, and CBA achieve lower rates (e.g., 32.9% and 39.3%). This indicates that placing the patch beneath the target is more effective in exploiting model vulnerabilities.
2) *Observation 2:* Compared to methods that place the patch beneath the target (BADEI and Adv Texture), our method shows comparable or slightly better performance in white-box attacks. However, our method demonstrates better black-box attack transferability. For example, on YOLOv5x, our method achieves a 51.4% success rate in black-box attacks, while Adv Texture and BADEI

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Fig. 3. Visual examples of applying adversarial patches. APPA(out) means that the patch is placed on and out of the target.

TABLE II

ASR (%) OF THE ATTACK METHODS IN CONDUCTING ADVERSARIAL BACKGROUND PATCH ON VICTIM MODELS, USING THE DOTA AND RSOD DATASETS. THE WHITE-BOX ATTACK IS BOLDED, AND THE OTHERS ARE BLACK-BOX ATTACKS. THE *Y* STANDS FOR THE YOLO OBJECT DETECTION MODEL, WITH THE FOLLOWING NUMBERS AND LETTERS REPRESENTING ITS VERSION AND SIZE, RESPECTIVELY

Datasets	Patches\Detectors	Faster R-CNN [50]	SSD [51]	*Y*3n [52]	*Y*5n [53]	*Y*5s [53]	*Y*5m [53]	*Y*5l [53]	*Y*5x [53]	*Y*8n [54]	DETR [11]
DOTA	Faster R-CNN [50]	**56.45**	93.46	79.53	93.53	100	83.49	56.42	46.29	89.73	76.77
	SSD [51]	44.76	**92.35**	84.57	93.63	96.08	65.35	56.74	54.34	84.65	75.43
	*Y*3n [52]	15.36	71.77	**62.83**	75.66	67.94	48.75	32.00	12.85	67.88	66.92
	*Y*5n [53]	16.56	84.65	94.86	**85.94**	75.59	70.47	71.79	67.94	69.22	77.82
	*Y*5s [53]	79.55	93.61	100	100	**100**	100	98.72	100	100	83.37
	*Y*5m [53]	53.80	91.03	92.34	100	100	**92.34**	87.17	89.70	97.35	79.71
	*Y*5l [53]	41.04	94.96	93.55	82.07	88.50	74.34	**67.88**	64.08	94.71	68.37
	*Y*5x [53]	32.08	79.48	84.73	89.70	94.82	74.39	66.75	**82.05**	91.10	68.52
	*Y*8n [54]	15.37	89.73	83.29	84.62	96.17	65.40	44.75	26.89	**89.00**	64.22
	DETR [11]	37.16	83.30	87.24	85.88	83.37	84.56	53.76	69.27	88.56	**78.54**
RSOD	Faster R-CNN [50]	**51.27**	66.55	56.06	88.69	85.91	84.13	28.02	32.72	67.17	59.36
	SSD [51]	33.91	**70.49**	80.27	84.06	87.79	79.38	41.06	37.29	73.80	69.17
	*Y*3n [52]	34.55	25.57	**58.74**	44.88	49.56	28.05	29.31	29.29	35.47	48.74
	*Y*5n [53]	23.42	51.23	47.56	**50.50**	60.74	46.71	31.66	31.84	14.89	45.14
	*Y*5s [53]	58.95	84.60	94.45	97.21	**95.31**	94.49	80.25	85.18	88.72	51.21
	*Y*5m [53]	43.17	73.04	68.16	91.58	92.48	**90.62**	65.39	55.12	70.04	50.62
	*Y*5l [53]	39.81	70.49	80.32	84.14	88.74	79.38	**41.13**	37.31	73.77	69.02
	*Y*5x [53]	21.20	39.67	59.83	79.43	85.00	66.31	40.15	**51.43**	65.36	56.32
	*Y*8n [54]	11.25	60.24	82.27	87.82	91.47	49.46	14.89	14.90	**59.83**	52.37
	DETR [11]	39.92	41.01	52.30	60.73	52.32	72.94	22.38	42.14	43.90	**63.85**

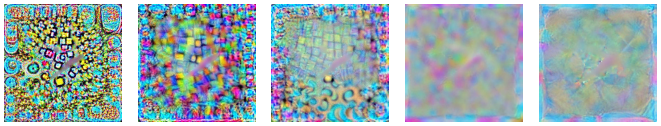


Fig. 4. Adversarial background patch generated using different TV smooth loss weight α.

achieve 37.3% and 31.6%, respectively. This indicates that our method can better generalize across different models, making it more effective in practical scenarios where the target model's architecture is unknown.

E. Physical Attack

In this section, we do a comprehensive experiment to verify the effectiveness of the adversarial background patch in the physical world.

1) Impact of TV Smoothness Loss: In order to study the effect of TV smooth Loss weight α, for the YOLOv5l model, different alpha values are used to generate the corresponding

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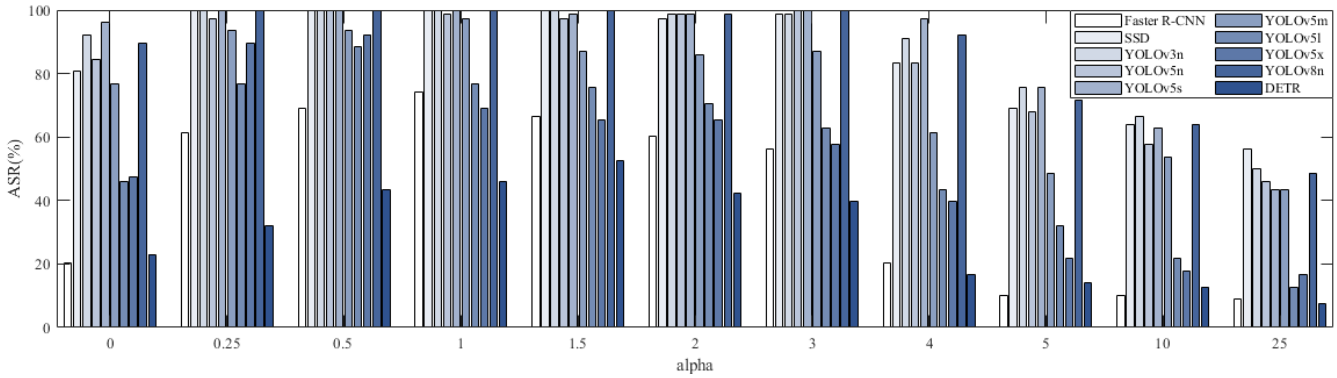


Fig. 5. ASR (%) of the attack methods in conducting adversarial background patch on victim models with different TV smooth loss weight α.

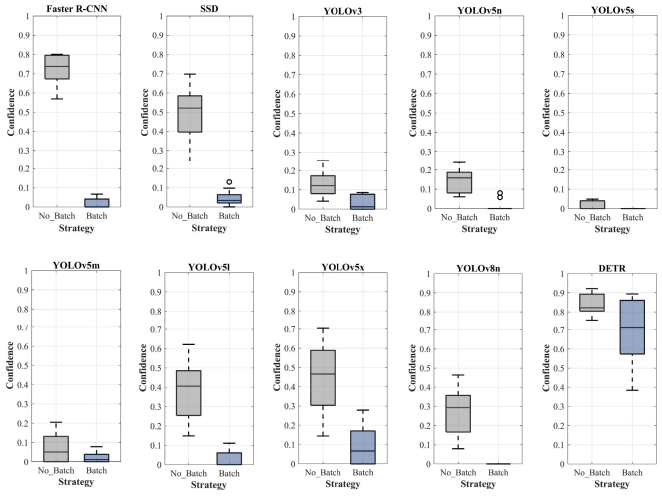


Fig. 6. Average confidence of the protected object during the attack when using and not using batch optimization in the adversarial background patch generation process.

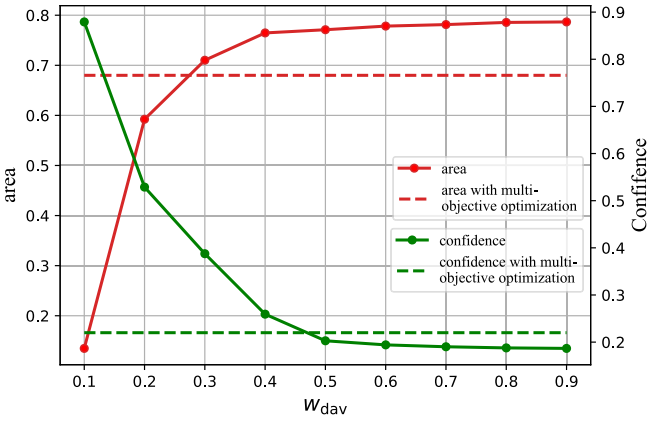


Fig. 7. Area size and objective confidence of the adversarial background patches generated using different $w_{adv}(w_{area} = 1 - w_{adv})$, compared to the adversarial background patches generated using multiobjective optimization.

adversarial background patch, and the results are shown in Fig. 4. When alpha is 0, the fine-grained image is higher, and the details are richer. As the alpha value increases, some color transitions and pattern structure changes become more and more obvious, such as when alpha is 2.5. However, when alpha is 5, the image texture information begins, and only simple color blocks can be observed in the end. In order to verify the impact of these changes on the attack effect, we use YOLOv5n for white-box and others for black-box testing, respectively. The ASR is shown in Fig. 5. When alpha is small, the ASR increases to a certain extent with the increase of alpha, which may be due to the increase of the adversarial information captured by the victim model with the appearance of texture information. When alpha exceeds a certain value, the loss of texture information becomes more and more serious, and the ASR is significantly reduced.

2) Effectiveness of Batch Optimization: For the ablation experiment of batch optimization, according to the above settings, only the batch optimization step is removed from the method. We use the YOLOv5n model as an attack object to make patches. Besides, two sizes of aircraft models, 16 and 20 cm, are used, and the confidence of ten victim models for each aircraft is recorded. In order to effectively display and analyze the distribution characteristics of the collected data, the recorded confidences are visualized by box plots. The experimental results are shown in Fig. 6. After using batch optimization, the success rate of black-box and white-box attacks has been improved to varying degrees, except that DETR has a slight decline. Specifically, the white-box attack remains 100%. The use of this strategy in the black-box attack reduces the confidence of SSD and DETR in objects by about 50%, and the confidence of other models has decreased by different degrees.

3) Effectiveness of Multiobjective Optimization Based on Pareto Efficiency: To demonstrate the effectiveness of the Pareto-based multiobjective optimization algorithm, we experimentally compare manual weight adjustment and Pareto optimization strategies, which are shown in Fig. 7. In the manual weight group, where the adversarial loss weight (w_{adv}) ranges from 0.1 to 0.9 and the area loss weight is $1 - w_{adv}$, the generated patches exhibit a clear trade-off dilemma. When $w_{adv} \geq 0.5$, the area ratio increases to above 0.75, but the average confidence drops below 0.3. Conversely, when $w_{adv} \leq 0.3$, the area ratio is compressed below 0.7, yet the confidence surges above 0.4, which means that the attack fails. Intermediate weights ($w_{adv} = 0.4$) partially balance the two objectives but fail to achieve simultaneous minimization.

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TABLE III

ASR (%) OF THE ATTACK METHODS IN CONDUCTING ADVERSARIAL BACKGROUND PATCH ON VICTIM MODELS, USING THE RSOD DATASETS. THE WHITE-BOX ATTACK IS BOLDED, AND THE OTHERS ARE BLACK-BOX ATTACKS. THE *Y* STANDS FOR THE YOLO OBJECT DETECTION MODEL, WITH THE FOLLOWING NUMBERS AND LETTERS REPRESENTING ITS VERSION AND SIZE, RESPECTIVELY

		Faster R-CNN [50]	SSD [51]	*Y*3n [52]	*Y*5n [53]	*Y*5s [53]	*Y*5m [53]	*Y*5l [53]	*Y*5x [53]	*Y*8n [54]	DETR [11]
Thys et al.	Faster R-CNN [50]	**53.1**	61.3	31.9	22.4	26.1	22.3	19.8	18.1	30.3	15.2
	SSD [51]	65.4	**75.2**	33.2	22.6	33.0	24.7	25.7	23.6	32.4	20.7
	YOLOv3n [52]	35.6	53.9	**36.9**	19.7	28.5	20.4	23.2	21.6	29.8	13.7
	YOLOv5n [53]	73.4	66.2	29.6	**22.8**	33.2	19.0	23.2	21.0	30.2	14.5
	YOLOv5s [53]	39.2	54.5	33.3	20.2	**32.9**	24.0	23.9	22.5	31.4	15.7
	YOLOv5m [53]	38.0	52.8	30.4	19.4	29.2	**24.4**	22.7	19.8	52.8	15.2
	YOLOv5l [53]	35.5	50.8	31.8	19.0	29.9	23.4	**24.7**	21.3	32.6	13.8
	YOLOv5x [53]	35.1	55.5	30.5	18.1	30.3	22.5	23.2	**23.1**	29.5	16.2
	YOLOv8n [54]	65.4	75.2	33.2	22.6	33.0	24.7	25.7	23.6	**36.6**	20.7
	DETR [11]	43.2	39.2	7.4	10.1	9.4	6.5	6.3	4.8	29.9	**18.0**
APPA (outside)	Faster R-CNN [50]	**69.7**	50.6	27.8	43.3	29.7	28.0	24.8	25.8	37.5	42.3
	SSD [51]	57.4	**55.4**	31.5	45.9	34.0	37.1	28.7	32.9	33.6	37.1
	YOLOv3n [52]	53.6	50.6	**35.5**	43.0	33.1	32.6	27.2	31.9	35.2	42.1
	YOLOv5n [53]	59.1	53.4	35.6	**60.6**	33.9	33.4	30.4	32.5	38.8	33.1
	YOLOv5s [53]	65.4	52.0	31.3	41.9	**39.3**	29.6	26.4	28.5	37.8	40.6
	YOLOv5m [53]	58.8	58.9	36.1	48.2	46.4	**44.8**	37.3	38.5	29.6	36.4
	YOLOv5l [53]	66.0	57.4	32.9	42.0	42.6	36.1	**45.8**	36.1	35.2	36.9
	YOLOv5x [53]	59.5	51.4	34.3	42.1	38.7	35.6	31.0	**37.3**	32.0	34.9
	YOLOv8n [54]	57.4	55.4	31.5	45.9	34.0	37.1	28.7	32.9	**32.1**	37.1
	DETR [11]	53.0	48.2	27.9	38.2	28.8	30.1	27.4	33.6	48.2	**42.1**
CBA	Faster R-CNN [50]	**71.2**	53.7	28.1	33.9	26.4	29.6	24.2	26.0	29.0	35.2
	SSD [51]	65.9	**60.3**	30.8	37.2	29.1	29.8	24.9	25.9	33.1	39.0
	YOLOv3n [52]	62.0	45.3	**24.6**	27.0	23.1	23.7	21.8	22.4	30.9	33.7
	YOLOv5n [53]	68.9	52.8	35.1	**49.2**	34.8	37.4	32.8	30.9	28.9	37.8
	YOLOv5s [53]	65.9	50.6	28.2	36.9	**31.4**	31.4	25.7	25.7	29.9	33.3
	YOLOv5m [53]	63.3	48.1	26.3	31.5	27.5	**31.7**	25.3	24.3	33.2	34.4
	YOLOv5l [53]	61.1	49.2	27.4	34.9	28.5	29.7	**26.7**	25.3	27.9	33.2
	YOLOv5x [53]	64.3	49.3	30.0	39.6	31.6	34.8	29.8	**31.8**	29.5	33.5
	YOLOv8n [54]	65.9	60.3	30.8	37.2	29.1	29.8	24.9	25.9	**36.5**	39.0
	DETR [11]	70.0	49.4	24.2	27.7	25.4	27.4	23.8	23.5	25.9	**39.6**
BADEI	Faster R-CNN [50]	**81.9**	77.2	77.2	80.3	90.6	70.9	40.2	37.8	60.6	38.7
	SSD [51]	22.8	**76.4**	78.0	71.7	89.8	47.2	14.2	31.5	62.2	39.4
	YOLOv3n [52]	13.4	22.0	**37.0**	35.4	39.4	44.1	18.9	21.3	33.1	34.7
	YOLOv5n [53]	12.6	22.0	40.2	**59.1**	40.2	53.5	16.5	22.8	47.2	41.0
	YOLOv5s [53]	15.0	23.6	44.1	58.3	**55.9**	40.2	19.7	18.1	38.6	48.9
	YOLOv5m [53]	14.2	20.5	48.8	46.5	44.1	**49.6**	14.2	29.9	40.2	44.2
	YOLOv5l [53]	13.4	30.7	52.8	57.5	68.5	63.0	**44.9**	55.1	50.4	49.7
	YOLOv5x [53]	15.7	23.6	37.8	59.1	58.3	48.8	19.7	**44.9**	48.0	45.0
	YOLOv8n [54]	18.9	18.9	62.2	59.8	55.1	72.4	29.9	42.5	**61.4**	41.0
	DETR [11]	18.9	27.6	43.3	54.3	32.3	66.1	26.0	34.6	52.8	**54.4**
Adv Texture	Faster R-CNN [50]	**90.2**	59.3	47.2	64.5	63.6	53.3	31.2	22.2	51.9	21.2
	SSD [51]	24.8	**84.2**	30.2	41.3	43.7	30.3	14.4	10.8	41.2	23.1
	YOLOv3n [52]	23.1	19.2	**61.0**	33.3	37.2	38.5	27.3	21.8	31.1	38.4
	YOLOv5n [53]	34.3	20.9	33.2	**71.4**	35.7	58.6	28.9	23.8	39.7	24.5
	YOLOv5s [53]	28.1	24.8	46.3	50.5	**94.2**	67.2	37.4	31.6	47.2	21.1
	YOLOv5m [53]	24.2	18.2	43.4	54.6	48.3	**91.5**	33.3	24.5	53.9	20.5
	YOLOv5l [53]	16.5	15.6	40.1	46.8	42.2	69.4	**73.5**	28.1	49.1	29.1
	YOLOv5x [53]	17.3	17.1	35.5	51.2	39.1	58.3	30.8	**65.3**	48.3	26.3
	YOLOv8n [54]	12.2	24.5	59.8	78.0	68.2	41.3	25.3	23.3	**77.2**	38.3
	DETR [11]	30.2	10.6	40.6	42.3	38.3	47.9	25.9	10.9	34.7	**83.9**
Ours	Faster R-CNN [50]	**51.2**	66.5	56.0	88.6	85.9	84.1	28.0	32.7	67.1	59.3
	SSD [51]	33.9	**70.4**	80.2	84.0	87.7	79.3	41.0	37.2	73.8	69.1
	YOLOv3n [52]	34.5	25.5	**58.7**	44.8	49.5	28.0	29.3	29.2	35.4	48.7
	YOLOv5n [53]	23.4	51.2	47.5	**50.5**	60.7	46.7	31.6	31.8	14.8	45.1
	YOLOv5s [53]	58.9	84.6	94.4	97.2	**95.3**	94.4	80.2	85.1	88.7	51.2
	YOLOv5m [53]	43.1	73.0	68.1	91.5	92.4	**90.6**	65.3	55.1	70.0	50.6
	YOLOv5l [53]	39.8	70.4	80.3	84.1	88.7	79.3	**41.1**	37.3	73.7	69.0
	YOLOv5x [53]	21.2	39.6	59.8	79.4	85.0	66.3	40.1	**51.4**	65.3	56.3
	YOLOv8n [54]	11.2	60.2	82.2	87.8	91.4	49.4	14.8	14.9	**59.8**	52.3
	DETR [11]	39.9	41.0	52.3	60.7	52.3	72.9	22.3	42.1	43.9	**63.8**

In contrast, the multiobjective optimization based on the Pareto efficiency method adaptively allocates weights without manual intervention, achieving superior equilibrium: the optimized patches attain an area ratio of 0.68 and an average confidence of 0.23, outperforming the best manual trade-off.

4) Verification of Patch Occlusion Robustness: In this section, to verify the effectiveness of the adversarial background patch attack under different occlusion conditions, we place aircraft models at various angles on adversarial background patches of different sizes, and conduct experiments using different object detectors.

Specifically, we use aircraft models of 16 cm, and rotate them from 0° to 45°, 90°, 135°, 180°, 225°, 270°, and 315°, respectively, and the side length of the adversarial background patch is set to 0.4, 0.6, 0.8, 1, and 1.2 times the length of the aircraft. As shown in Fig. 8, in the implementation process, we use the adversarial background patch generated by YOLOv5n, because its attack effect is better in the pre-experiment. At the same time, we use all ten object detectors for testing to verify their transferability. The IoU threshold is set to 0.2, that is, the detection result with an IoU value below 0.2 is recorded as 0. The experimental results are shown in Table IV. Here are the key observations.

1) *Observation 1:* Under the condition of white-box attack, the adversarial background patch achieves complete hiding of objects under different occlusion conditions and

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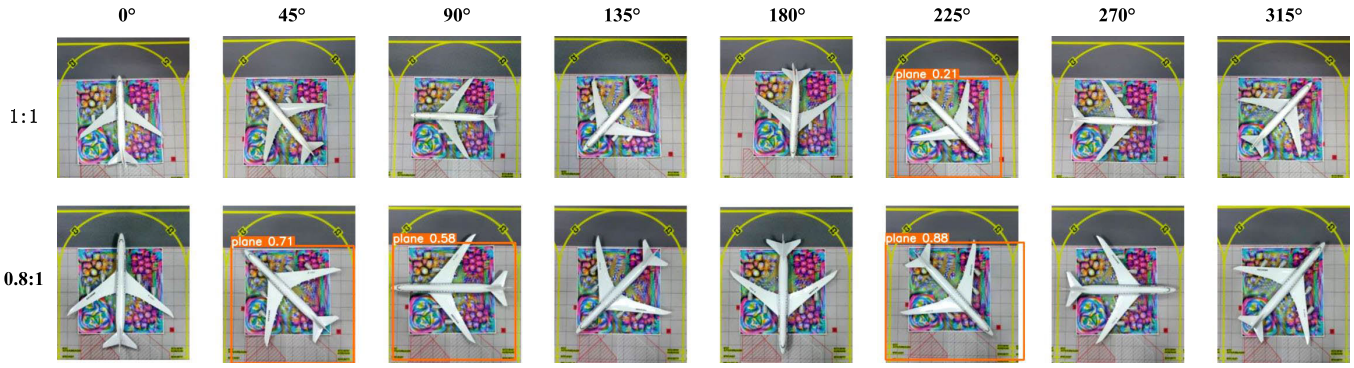


Fig. 8. Examples of physical attacks using 1:1 and 0.8:1 adversarial background patches, specifically, using an adversarial background patch generated for the YOLOv5n model and placing the model at eight separate angles.

TABLE IV

AVERAGE CONFIDENCE OF DIFFERENT OBJECT DETECTORS FOR EACH TARGET IN A PHYSICAL ATTACK, YOLOV5N IS A WHITE-BOX ATTACK, AND THE REMAINING MODELS ARE BLACK-BOX ATTACKS

Detector\size	**0.4:1**	**0.6:1**	**0.8:1**	**1:1**	**1.2:1**
Faster R-CNN	0.828	0.54	0	0	0.094
SSD	0.642	0.324	0.234	0.280	0.238
YOLOv3n	0.114	0.138	0	0	0
YOLOv5n	0.314	0.214	0	0	0
YOLOv5s	0.278	0	0	0	0
YOLOv5m	0.848	0.364	0	0	0
YOLOv5l	0.908	0.900	0	0	0.064
YOLOv5x	0.924	0.526	0.052	0	0
YOLOv8n	0.440	0.110	0	0	0
DETR	0.968	0.830	0.840	0.640	0.263

TABLE V

ASR(%) OF ADVERSARIAL BACKGROUND PATCHES ON DIFFERENT OBJECT DETECTORS AT DIFFERENT SIZES

Size	Method	Faster R-CNN	YOLOv5s	YOLOv5l
1.5:1	CBA [28]	-	-	-
	BADEI [29]	98.76	**100.00**	**100.00**
	Adv Texture [30]	98.65	98.29	91.87
	Ours	**100.00**	**100.00**	99.52
1:1	CBA [28]	**98.89**	**100.00**	**99.45**
	BADEI [29]	96.91	99.58	98.82
	Adv Texture [30]	60.14	53.61	47.51
	Ours	93.51	**100.00**	98.32
0.8:1	CBA [28]	-	-	-
	BADEI [29]	78.34	82.64	80.90
	Adv Texture [30]	25.26	30.11	22.35
	Ours	**96.45**	**98.86**	**93.33**

sizes when its size is 0.8:1 or larger. At a size of 0.6:1, it still maintains strong attack effectiveness, reducing the average confidence score to 0.214. The attack effect drops significantly at a size of 0.4:1. As shown in Table V, our adversarial background patch can maintain a stronger attack effect at 1 and 0.8 times the size of the object compared to methods such as CBA, Adv Texture, and BADEI.

2) *Observation 2:* In the black-box attack, when the adversarial background patch has a size ratio greater than 0.8:1, it achieves superior attack performance on object detectors other than DETR. When its size is 0.6:1, it still maintains good attack performance on smaller-scale object detection models such as YOLOv5s, YOLOv5m, YOLOv3n, and YOLOv8n, but the attack performance drops significantly for larger-scale models like YOLOv5l and YOLOv5x. When the size decreases to 0.4:1, due to excessive occlusion, it loses its attack effectiveness for most models. For DETR, possibly due to the unique structure of the transformer, the adversarial background patch has the poorest attack effect, but it can still achieve some attack effect at sizes of 1:1 and 1.2:1.

5) Physical Implementation Results: We printed the adversarial background patch generated by different models and tested them with aircraft models. During the test, various object detection models were used to detect all experimental images to verify the transferability of the adversarial background patch, and the confidence threshold is set to 0.2. The test results are shown in Figs. 9 and 10. Here are the key observations.

1) *Observation 1:* In the physical world, the adversarial background patch shows strong attack ability and good attack mobility. Specifically, the adversarial background patch makes the detection success rate of the detection model less than 25%, and the confidence of the detection object is basically less than 0.8. In terms of attack versatility, the adversarial background patches generated by different models show strong attack capabilities against other models, especially between the YOLO series. It is worth mentioning that YOLOv5x and YOLOv5l show strong robustness, which may be due to the large size of the model.

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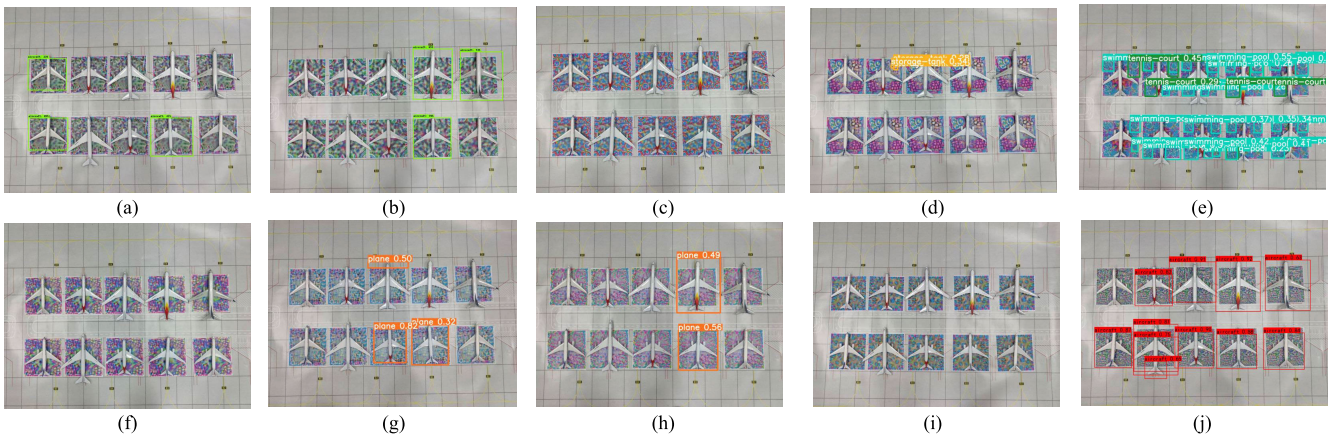


Fig. 9. Results of a white-box attack on ten object detectors in the physical world: (a) Faster R-CNN, (b) SSD, (c) YOLOv3n, (d) YOLOv5n, (e) YOLOv5s, (f) YOLOv5m, (g) YOLOv5l, (h) YOLOv5x, (i) YOLOv8n, and (j) DETR.

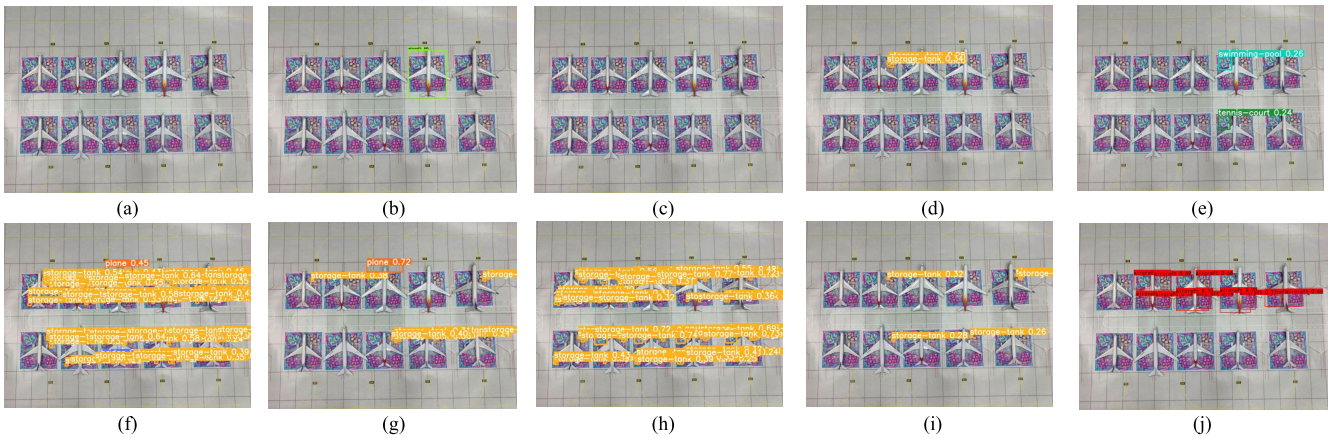


Fig. 10. Results of a black-box attack on ten object detectors in the physical world. The patch used is based on YOLOv5n: (a) Faster R-CNN, (b) SSD, (c) YOLOv3n, (d) YOLOv5n, (e) YOLOv5s, (f) YOLOv5m, (g) YOLOv5l, (h) YOLOv5x, (i) YOLOv8n, and (j) DETR.

2) *Observation 2:* By observing the detected objects, we have the following observation: 1) in white-box attacks, DETR proves to be more difficult to attack. This may be due to the higher fine-grained detail of the adversarial background patch obtained through optimization, which makes it harder to form obvious texture information that can be captured by optical sensors. As a result, the attack effectiveness of the adversarial background patch generated for DETR can be even lower than that produced using YOLOv5n in black-box attacks and 2) the texture information in the adversarial background patch has a high likelihood of being recognized as another class by the object detector, which reduces the chances of the protected object being detected. This may be due to the fact that in the adversarial loss, we only reduce the confidence score of the protected object.

6) Experiments on Attack Range: To verify the attack range of our proposed adversarial background patches, we conducted experiments by moving the patches in four directions (up, down, left, and right) relative to the target object. The movement distances were set as fractions of the background side length s, specifically $s/4$, $s/2$, and s. We recorded the average confidence of ten target objects under different movement conditions. The experimental results are presented in Table VI. The results show that the adversarial background patches maintained a high level of effectiveness across different movement conditions and models. For instance, in the case of the YOLOv5l model, the average confidence of the target objects remained relatively low even when the target was moved by a distance of s in any direction. Similarly, other models such as Faster R-CNN and DETR still maintain a low average confidence when the object moves $s/4$, $s/2$ distance relative to the adversarial background patch.

7) Physical Attack Robustness: To verify the robustness of our proposed adversarial background patches for physical attacks, we simulated lighting changes and post-processing operations of optical sensors in the physical world by adjusting brightness and contrast. In addition, we further validated their robustness to shooting angles by changing the camera's shooting angle. The experimental results are shown in Fig. 11. We found that our adversarial background patches successfully protected all targets from being recognized by the object detector under varying lighting, contrast, and angle conditions. This indicates that our method has strong robustness in the dynamic physical environment.

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TABLE VI
EXPERIMENTS ON THE ATTACK RANGE OF ADVERSARIAL BACKGROUND PATCHES

Model\Location	0	Up			Down			Left			Right		
		s/4	*s*/2	*s*	*s*/4	*s*/2	*s*	*s*/4	*s*/2	*s*	*s*/4	*s*/2	*s*
Faster-RCNN	0.000	0.000	0.117	0.756	0.000	0.224	0.681	0.000	0.162	0.746	0.000	0.199	0.589
SSD	0.049	0.023	0.028	0.214	0.087	0.135	0.355	0.022	0.077	0.267	0.020	0.077	0.317
YOLOv3	0.052	0.000	0.000	0.257	0.000	0.124	0.255	0.000	0.092	0.104	0.000	0.171	0.285
YOLOv5n	0.000	0.000	0.000	0.072	0.000	0.000	0.058	0.000	0.000	0.036	0.000	0.035	0.142
YOLOv5s	0.000	0.000	0.000	0.136	0.114	0.172	0.195	0.000	0.127	0.172	0.075	0.157	0.356
YOLOv5m	0.000	0.099	0.241	0.597	0.061	0.260	0.681	0.117	0.229	0.475	0.115	0.33	0.464
YOLOv5l	0.089	0.091	0.402	0.698	0.223	0.509	0.732	0.271	0.372	0.527	0.245	0.364	0.454
YOLOv5x	0.000	0.079	0.166	0.782	0.182	0.357	0.642	0.086	0.156	0.729	0.231	0.391	0.590
YOLOv8	0.000	0.000	0.000	0.024	0.000	0.000	0.047	0.000	0.102	0.158	0.000	0.083	0.261
DETR	0.199	0.208	0.450	0.639	0.235	0.533	0.722	0.246	0.492	0.737	0.158	0.374	0.762

* 0 means object has not moved.
* The *s* refers to the side length of the adversarial background patch.

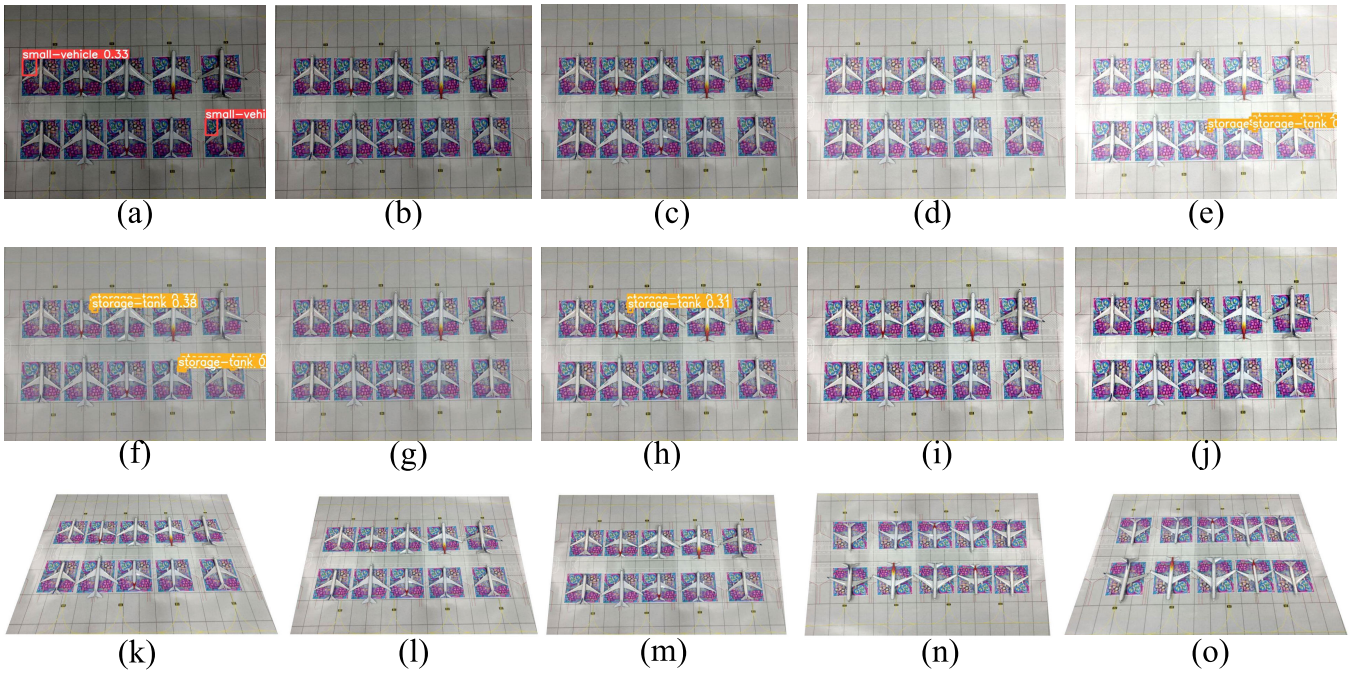


Fig. 11. Physical attack robustness experiments. Demonstrate the attack effect under different (a)–(e) lighting conditions, (f)–(j) contrasts, and (k)–(o) shooting angles.

TABLE VII
EXPERIMENTS COMPARING THE COST OF TIME AND THE COST OF COMPUTING RESOURCES

Method	ASR (%)	Time Cost		GPU Calculate Cost	
		iter(s)	total(s)	Mem(%)	Usage(%)
Thys et. al [58]	32.9	1.53	734	**14.31**	**18.69**
APPA(out) [27]	39.3	3.35	1780	32.49	35.77
CBA [28]	31.4	1.72	845	28.63	41.57
BADEI [29]	87.5	1.84	1032	18.00	20.19
Adv Texture [30]	94.2	2.32	1243	67.32	72.43
Ours	**95.3**	**1.32**	**632**	40.06	47.39

* The experiment is conducted with NVIDIA GeForce RTX 4090 GPU (24GB VRAM), AMD Ryzen 9 7950X3D and 64GB DDR5 RAM.

F. Attack Cost

In this section, we conducted a comprehensive evaluation of the attack cost, which includes both time cost and GPU computational resources. Specifically, we rigorously timed the single iteration duration and the total time required to generate adversarial background patches, and monitored GPU memory usage and GPU utilization. The results of the experiment are shown in Table VII. Here are the key observations.

1) *Observation 1:* Considering time cost, our method stands out by requiring a shorter single iteration time and total time for generating adversarial background patches. For example, compared to the method by Thys et al. [58], our method reduces the single iteration time from 1.53 to 1.32 s, and the total time from 734 to 632 s. Similarly, when compared to other methods, our method consistently shows a reduction in both single iteration and total time. This efficiency is crucial in practical scenarios where time is a limiting factor for generating and applying adversarial background patches.

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2) *Observation 2:* Considering GPU computational cost, our method exhibits lower GPU memory usage and computational resource usage in comparison to other methods. While Thys et al. [58] have the lowest GPU memory usage at 14.31% and computational resource usage at 18.69%, our method achieves a higher adversarial success rate of 96.3% while maintaining GPU memory usage at 40.06% and computational resource usage at 47.39%. This balance between resource usage and effectiveness is not observed in other methods like Adv Texture [30], which, despite having a higher adversarial success rate of 94.2%, consumes a significantly higher GPU memory of 67.32% and computational resources of 72.43%. This makes our method more suitable for resource-constrained environments, which is a key factor in ensuring its adaptability to various scenarios.

V. Conclusion

In this article, we propose adversarial background patches. By considering the occlusion robustness of the adversarial background patch and using a batch optimization method based on the attack difficulty, the generated adversarial background patch has strong occlusion robustness. In addition, the proposed area loss balances the attack effect of the adversarial background patch and the patch area size, which reduces the economic cost of physical implementation. Finally, a multi-objective optimization method based on Pareto efficiency is introduced to solve the problem of the conflict between the adversarial loss and the area loss in the optimization process. Experimental results show that the adversarial background patches generated for object detectors with different structures have strong attack performance and portability in both the digital domain and the physical world. In addition, the use of area loss enables the adversarial background patches to achieve the adversarial effect with a smaller area.

However, the attack effectiveness of the adversarial background patches generated in this article for certain object detection still needs to be improved. In addition, the method based on iterative optimization makes it slow to generate an adversarial background patch. Looking ahead, there are two research directions. On the one hand, we consider the use of generative modeling to improve the generation speed of adversarial background patches. On the other hand, while maintaining the effectiveness against background attacks, we can explore how to improve texture concealment by using methods such as style transfer.

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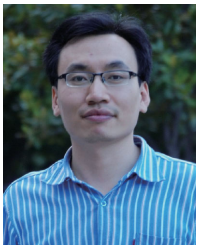
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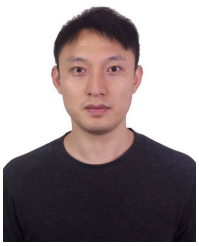
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